

Problem Understanding and Framing

Predicting university student dropout risk for targeted, supportive intervention

Domain	Education - student success and retention
Decision point	End of the first semester
Task type	Supervised multiclass classification with a dropout-risk score
Primary user	Student support and academic advising teams

This capstone frames student dropout as an early-warning decision problem. The proposed model estimates each student's probability of the Dropout outcome after first-semester results become available. The score is used to prioritize limited outreach capacity for counseling, tutoring, academic advising, and financial-support triage. It is not used to deny services, discipline students, or make an irreversible automated decision.

The modeling objective reflects the higher cost of missing a student who may need support while keeping the alert list usable. The numeric thresholds in this report are explicit proof-of-concept criteria, not claims about a university's actual staffing capacity. They are fixed before model selection and can be replaced with locally approved values during a pilot.

1. Business problem and decision

Universities often have limited staff and resources for student-retention programs. When support is offered only after a student has disengaged, the opportunity for a timely intervention may already have narrowed. The operational question is therefore: which students should receive proactive, human-reviewed support after the first semester?

Element	Definition
Decision	Rank students by estimated dropout risk and create a manageable outreach list.
Action	Offer supportive contact, needs assessment, advising, tutoring, or financial-aid referral.
Beneficiaries	Students who may benefit from earlier assistance; support teams that need consistent prioritization.
Planning scenario	Assume staff can review 30% of the eligible cohort. This is an illustrative capstone capacity, not a measured institutional limit.
Safeguard	A trained staff member reviews alerts; the score never triggers an adverse action on its own.

2. Data science formulation

The analytical task is supervised multiclass classification with three outcomes: Dropout, Enrolled, and Graduate. Although all three probabilities are produced, the operational signal is $P(\text{Dropout})$, which provides a continuous risk ranking for triage.

Component	Specification	Rationale
Unit of analysis	One student record	Matches the case reviewed by support staff.
Prediction point	End of first semester	Balances early action with the availability of meaningful academic signals.
Target	Student outcome: Dropout, Enrolled, or Graduate	Preserves the distinction between continuing and completed students.
Operational output	Probability of Dropout plus risk rank	Supports capacity-based prioritization instead of a rigid yes/no label.
Primary error concern	False negatives	A missed at-risk student loses the opportunity for early support.
Secondary error concern	False positives	Unnecessary outreach consumes capacity and may stigmatize a student if handled poorly.

3. Operational workflow

1. At the end of the first semester, validated administrative and academic records are assembled using only information available at that time.
2. The model generates class probabilities; $P(\text{Dropout})$ becomes the prioritization score.
3. For the proof-of-concept scenario, the highest-risk 30% form the review queue. This share is close to the dataset's 32.12% dropout prevalence and tests whether the model can concentrate most eventual dropouts into a smaller, manageable group. An institution should replace 30% with its verified service capacity.
4. Authorized staff review contextual information, contact the student, and offer voluntary support matched to need.
5. Outcomes, outreach uptake, false-positive burden, and subgroup performance are monitored for continuous improvement.

Stakeholders and accountability

Student-support leadership owns the operational decision; academic advisers, counseling services, tutors, and financial-aid staff deliver interventions. Data and governance teams are responsible for data quality, access control, fairness monitoring, and model review. Students remain affected stakeholders and must be treated with dignity, transparency, and an avenue for questions or correction.

4. Success criteria

Model quality and operational usefulness are evaluated together. The four numeric targets below are project design assumptions chosen before model selection. They make the evaluation transparent and interpretable; they are not published sector benchmarks.

Measure	Project target	Basis for the number
Dropout recall	$\geq 75\%$	Requires the model to identify at least three of every four eventual dropouts, because missed cases lose the opportunity for early support.
Dropout precision	$\geq 50\%$	Requires at least one eventual dropout for every two alerts, limiting unnecessary contact and staff workload.
Primary selection metric	Dropout F2 score	Gives recall greater influence than precision because a false negative is treated as the more costly error.
Review capacity	Top 30% of risk ranking	Illustrative scenario near the dataset's 32.12% dropout prevalence. It reduces routine review to about three students in ten and must be recalibrated to actual staffing.
Capacity capture	$\geq 60\%$ of actual dropouts	Requires the 30% queue to contain at least three of every five dropouts. A random 30% list would capture about 30%, so the target demands at least twice that concentration.

The 30% queue and the 60% capture target are evaluated together. The first controls workload; the second requires the ranking to add value within that workload. Actual deployment thresholds should be set jointly by student-support leadership and governance teams using staffing, intervention cost, student experience, and subgroup performance.

5. Business and service KPIs

- Coverage: share of eventual dropouts included in the capacity-limited review list.
- Workload: number and percentage of students sent for human review, by period and team.
- Engagement: outreach completion, successful contact, referral acceptance, and intervention uptake.
- Student outcome: semester-to-semester persistence and eventual completion among contacted students, compared through an appropriately designed pilot.
- Burden and safety: unsuccessful contacts, complaints, opt-outs, and the experience of students who were flagged but did not drop out.
- Equity: recall, false-positive rate, selection rate, and alert burden across audited student groups.

Retention improvement and financial return are not claimed from historical prediction alone. They require a prospective evaluation that records intervention costs, student participation, and incremental outcomes relative to a valid comparison group.

6. Scope, assumptions, and responsible-use boundaries

Area	Boundary or assumption
In scope	Risk prioritization after first-semester results using authorized institutional data.
Out of scope	Admissions, scholarships, grading, disciplinary action, or automatic withdrawal decisions.
Timing	Second-semester academic variables are excluded from the primary early-warning model to prevent information leakage.
Interpretation	Risk is not destiny; the score is an imperfect signal that requires contextual human review.
Privacy	Access should be role-based, data-minimized, logged, and retained only as policy permits.
Fairness	Sensitive and proxy variables require explicit review; subgroup performance must be monitored before and after deployment.
Generalisability	The source data represents one Portuguese higher-education institution; local validation is required before use elsewhere.

7. Traceability to later capstone tasks

The framing governs the rest of the project: Task 2 checks whether the public dataset supports the prediction point; Task 3 removes second-semester leakage and engineers first-semester features; Task 4 selects a model against the recall, precision, and capacity targets; Task 5 audits subgroup impacts; and Task 6 communicates the evidence and safeguards to technical and business audiences.

Reference

UCI Machine Learning Repository. [Predict Students' Dropout and Academic Success](#). DOI: 10.24432/C5MC89. Licensed under CC BY 4.0.